

PCCST501 – COMPUTER NETWORKS

ASSIGNMENT 4 – MODULE 4

SNMP, ASN.1, Physical Layer & Transmission Media

Course Outcome: CO4 **Module:** 4

Syllabus Coverage: SNMP, ASN.1; Data and signals; Digital transmission; Analog transmission; Bandwidth utilization; Transmission media.

Instructions: Answer all questions. Use neat diagrams wherever required. Answers should be technically precise and should follow the terminology used in the prescribed textbook.

Assignment Questions

1. Explain the purpose of **SNMP** in network management. Describe the roles of the SNMP manager, SNMP agent and managed device.
2. Explain the major SNMP operations used for retrieving and modifying management information. Distinguish between manager-initiated requests and notifications.
3. What is a **Management Information Base (MIB)**? Explain the role of Object Identifiers (OIDs) in SNMP with a suitable example.
4. Explain **ASN.1 (Abstract Syntax Notation One)**. Why is a standardized data description notation useful in network management?
5. Explain the relationship among **ASN.1, SNMP management information and data representation**. Illustrate the explanation with a simple structured example.
6. What are **data and signals**? Explain the characteristics of analog and digital signals, including amplitude, frequency, phase and signal period.
7. Explain **digital transmission** and **analog transmission**. Compare the two approaches with suitable examples and applications.
8. Explain **bandwidth utilization**. Discuss techniques used to improve the efficient use of available bandwidth, with suitable examples.
9. Compare **twisted-pair cable, coaxial cable and optical fiber** as transmission media. Consider bandwidth, attenuation, interference, distance, cost and applications.
10. Prepare a **transmission-media selection case study** for a college campus. Recommend suitable media for (i) a computer laboratory, (ii) an inter-building backbone and (iii) wireless access, and justify each selection based on the physical-layer characteristics studied in this module.

Expected Deliverables

- PDF report
- Relevant diagrams for SNMP/ASN.1 and signal concepts
- Transmission-media comparison table
- Campus transmission-media selection diagram
- References used

Evaluation Rubric

Criteria	Marks
Correctness and Technical Content	4
Diagrams / Analysis / Case Study	2
Documentation and Presentation	2
Viva / Demonstration	2
Total	10

Submission Format

Submit one PDF report through the course GitHub repository. The repository should contain a clear README.md and all diagrams/figures used in the assignment.